Listening Matrices and the Allocation of Guaranteed Wallets

Tonify Research

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1 Problem statement

The primary task is to generate a listening matrix satisfying specified hard constraints and statistical targets. On each accepted matrix we compare two deterministic rules allocating the same guaranteed budget: pro-rata (PR), which allocates the pooled budget in proportion to total artist plays, and user-centric (UC), which allocates each listener's wallet in proportion to that listener's plays.

PR and UC are alternative allocations of the same budget. Direct artist payments can coexist with either rule and are outside this initial study. Wallets are net amounts available for allocation after any funding fee; subscription tiers, service fees and label contracts are deferred. The full previous treatment is retained in `THEORY.v0.5.md`. No novelty claim is made here for the elementary identities or inequalities.

2 Conceptual formulation

A matrix cell measures either play counts or total listening duration for one artist–listener pair during one common observation period. Units must be consistent across the matrix. A generator may produce a random matrix; once realized, payouts and weighted means are finite deterministic calculations. Behavioral responses to changing the payment rule are not modeled.

Marginal activity distributions do not determine which listeners are connected to which artists. A useful generator must also expose targets for degrees, nonzero pairs, concentration and dependence. Matching a few statistics is not evidence that a matrix represents the actual music market.

3 Mathematical formulation

3.1 Notation

Symbols are repeated at their first use below.

Symbol	Definition
N, i	Number of artists and artist index $i \in \{1, \dots, N\}$.
U, u	Number of listeners and listener index $u \in \{1, \dots, U\}$.
p, p_{iu}	Nonnegative artist-by-listener matrix and one cell.
P_u, P_i	Listener total $\sum_i p_{iu} > 0$ and artist total $\sum_u p_{iu} \geq 0$.
$T, \bar{P}$	Total plays $T = \sum_u P_u$ and mean listener activity $\bar{P} = T/U$.
A_i, K_u	Number of listeners of artist i and artists of listener u.

W, R, W_u	Common positive net wallet, total budget $R = UW$, and individual wallets in the extension.
PR_i, UC_i	Artist payouts under the two rules.
w_u^i	Artist-specific normalized weights p_{iu}/P_i, defined for $P_i > 0$.
m_i, H_i	Weighted arithmetic and harmonic means of listener totals.
CV_i^2	Finite weighted squared coefficient of variation $\sum_u w_u^i (P_u - m_i)^2/m_i^2$.
c_u	Auxiliary component $1/P_u - U/T$ in the equality criterion.

Our i, u, N, U correspond to the review's a, l, A, L; L denotes listeners, not artists (a typographical error in the review).

3.2 Definitions and edge cases

Let N be the number of artists, U the number of listeners, and $p_{iu} \geq 0$ the activity of listener u on artist i in a common period. Rows are artists; columns are listeners. Legacy sim4 sparse exports use the opposite orientation and must be transposed. Define listener totals $P_u = \sum_i p_{iu} > 0$, artist totals $P_i = \sum_u p_{iu}$, total activity $T = \sum_u P_u = \sum_i P_i$, and mean listener activity $\bar{P} = T/U$. For a common positive net wallet W and budget $R = UW$,

$$PR_i = UW\frac{P_i}{T}, \qquad UC_i = W\sum_u \frac{p_{iu}}{P_u}.$$

Here PR_i and UC_i are artist i's payouts. These are deterministic functions of the matrix and wallet. If $P_i = 0$, both payouts vanish but their ratio is undefined. With audience size $A_i = \#\{u : p_{iu} > 0\}$, $UC_i \leq WA_i$ since every share $p_{iu}/P_u \leq 1$.

Budget balance follows from the definitions:

$$\sum_i PR_i = UW, \qquad \sum_i UC_i = W\sum_u \frac{\sum_i p_{iu}}{P_u} = UW.$$

Thus $\sum_i (UC_i - PR_i) = 0$; this is not a corollary of Theorem 1. For individual net wallets $W_u \geq 0$ with $R = \sum_u W_u > 0$, define $PR_i = RP_i/T$ and $UC_i = \sum_u W_u p_{iu}/P_u$. The same budget identity holds.

3.3 Scope and assumptions

A1: active listeners and consistent units. Every included listener has $P_u > 0$. The integer prototype uses $p_{iu} \in \mathbb{Z}_+$, hence $P_u \geq 1$, combining former A1 and A3. The algebra also permits real listening durations. Activity thresholds, silent payers ("deaf listeners") and routing their money to the service are deferred.

A2: common net wallets in the main comparison. Theorem 1 uses a common W. Individual wallets use the extension above. Declining service fees and superfan subscription tiers are deferred.

Former A4 and A7 are incorporated into the definition of a deterministic payout on one matrix. Former A5 concerns direct payments and is deferred. Service fees are already deducted; former A6 was a separate label-contract assumption, also outside this experiment. Former A8 does not prescribe a universal tail law: distribution families and their parameters are generator inputs. Condition (C) belongs only to the deferred tail lemma L3 in v0.5, not to the conceptual hypotheses of this experiment.

3.4 Finite arithmetic on one matrix

L1: payout ratio and UC gain conditions. Why this matters. For an accepted matrix, payout changes can be computed exactly. Each listener's weight is their share of a particular artist's plays. An unweighted average over people cannot replace this calculation.

Theorem 1. *Under A1–A2, for artist i with $P_i > 0$, let $w_u^i = p_{iu}/P_i$ be normalized nonnegative weights, $m_i = \sum_u w_u^i P_u$ the weighted arithmetic mean, and $H_i = (\sum_u w_u^i/P_u)^{-1}$ the weighted harmonic mean. Then*

$$\frac{UC_i}{PR_i} = \bar{P}\sum_u \frac{w_u^i}{P_u} = \frac{\bar{P}}{H_i} = \frac{\bar{P}}{m_i}\frac{m_i}{H_i}.$$

Proof. Dividing $W\sum_u p_{iu}/P_u$ by UWP_i/T and canceling common factors gives $(T/U)\sum_u (p_{iu}/P_i)/P_u$. Substitute the definitions. All operations are finite sums; no stochastic interpretation is required. □

A sufficient bound. Cauchy–Schwarz applied to $\sqrt{w_u^i P_u}$ and $\sqrt{w_u^i/P_u}$ gives

$$\left(\sum_u w_u^i P_u\right)\left(\sum_u \frac{w_u^i}{P_u}\right) \geq \left(\sum_u w_u^i\right)^2 = 1.$$

Hence $H_i \leq m_i$ and $UC_i/PR_i \geq \bar{P}/m_i$. Equality of the means holds exactly when P_u is constant on the positive-weight support. Therefore $m_i \leq \bar{P}$ is sufficient for $UC_i \geq PR_i$; $m_i < \bar{P}$ guarantees a strict gain. An *upper* bound on m_i helps verify this condition; for example $\max_{u:p_{iu}>0} P_u \leq \bar{P}$ suffices. A lower bound alone does not establish it.

(a) Equality. For a given artist, $UC_i = PR_i$ exactly when $\sum_u p_{iu}(1/P_u - U/T) = 0$. Simultaneous equality is $pc = 0$, with $c_u = 1/P_u - U/T$: c belongs to the kernel of p. Constant listener activity is sufficient but not necessary: $p = (10, 20)$ with one artist gives $UC_1 = PR_1 = 2W$.

(c) Exact crossover. $UC_i \geq PR_i$ if and only if $H_i \leq \bar{P}$; strict inequality corresponds to $H_i < \bar{P}$. The condition involving m_i is only sufficient. Labels (a) and (c) are retained for comparison with the review; former (b) is now part of the definitions.

(d) Interpretation. If $m_i = \bar{P}$, then $UC_i/PR_i = m_i/H_i \geq 1$, strictly so when listener totals differ on the positive-weight support. This describes a particular weighted sequence. Changing a matrix changes its weights, means and total activity simultaneously. It does not imply that increasing dispersion always increases UC income. The former "dispersion premium" claim is withdrawn.

The review's coefficient-of-variation bound. For the standard finite definition $CV_i^2 = \sum_u w_u^i (P_u - m_i)^2/m_i^2$, the proposed $m_i/H_i \geq 1 + CV_i^2$ is false. Take

$$p = \begin{pmatrix} 9 & 1 \\ 1 & 99 \end{pmatrix}.$$

Artist 1 has $P_u = (10, 100)$, $w^1 = (0.9, 0.1)$, $m_1 = 19$ and $H_1 = 1000/91$. Thus $m_1/H_1 = 1.729$, while $1 + CV_1^2 = 1090/361 \approx 3.01939$. The review omitted its CV definition; this counterexample applies to the definition explicitly stated here. Expanding the square instead yields the correct identity

$$\frac{m_i}{H_i} - 1 = \sum_u w_u^i \frac{(P_u - m_i)^2}{m_i P_u} \geq 0.$$

The denominators differ from those of CV_i^2. With unequal wallets, the general ratio is $UC_i/PR_i = (T/R)\sum_u w_u^i W_u/P_u$; the unadjusted condition $H_i \le \bar{P}$ no longer characterizes its sign.

What it says. The matrix, or its artist-specific weighted audience means, is needed to determine the winner. Niche status, audience size or heterogeneity alone is insufficient. Both payouts and their common budget are checked after accepting a matrix; the desired direction of payout redistribution is not an optimization target.

4 Constrained matrix generation

4.1 Inputs and targets

The standalone `sim4/constrained_gen.py` takes artist and listener counts, total integer activity, a common period, seed, iteration budget, search parameters and statistical targets. Optionally all listener totals are fixed. Every move preserves nonnegative integer cells, total activity and positive listener totals; fixed column totals are also preserved. Impossible hard budgets are rejected before search. Real durations are supported by the payout formulas, not by this integer optimizer.

Distribution targets specify a family, parameters, fixed comparison grid, tolerance, positive weight and provenance (measured, assumed or derived). Scalar targets specify a value instead of a CDF. Supported families are normal and lognormal (`mu`, `sigma`; log-scale for lognormal), Pareto (`shape`, `scale`), and beta (`alpha`, `beta`). Optional truncation precedes optional upward integer rounding.

Available distributions cover listener and artist totals, listener degrees, artist audiences, positive pair counts, positive artist-specific weights, artist weight concentration, listener preference concentration, and weighted arithmetic and harmonic audience activity. Scalar metrics are zero fraction, top-artist play share, artist-total Gini and the correlation of log audience size with log mean plays per audience member. Undefined correlations fail acceptance. Artist means and weight concentrations exclude zero-play artists; artist totals and audience sizes include them. Pooled positive weights do not prescribe a separate law for every artist. The full metric contract is in `sim4/CONSTRAINED.md`.

4.2 Objective and stochastic search

For target j, let $Z_j(p)$ be its finite sequence of values, F_j its specified CDF, G_j a fixed threshold grid and $\widehat{F}_j(x;p) = \#\{z \in Z_j(p) : z \le x\}/|Z_j(p)|$ its empirical CDF. Define

$$r_j(p) = \max_{x \in G_j} |\widehat{F}_j(x;p) - F_j(x)|, \qquad J(p) = \frac{\sum_j \lambda_j [r_j(p)/\varepsilon_j]^2}{\sum_j \lambda_j},$$

where $\varepsilon_j > 0$ is the tolerance and $\lambda_j > 0$ its weight. For a scalar target, r_j is its absolute error. Acceptance requires *every* residual to satisfy its tolerance, as well as every hard constraint. A small aggregate objective alone is insufficient.

The CDF error is a maximum over a declared grid, not over the entire real line. It is not a hypothesis test or a p-value: cells are dependent and the matrix is fitted to the same targets. Standard KS test interpretation is inapplicable (see SciPy's KS documentation).

Starting from a random feasible integer matrix, transfer a positive integer amount between two cells. A donor cannot become negative or leave its listener inactive. With fixed listener totals, transfers stay in the same column. Improvements are accepted; worsening moves are accepted with probability $\exp(-\Delta J/\theta_t)$, where $\theta_t > 0$ decreases over the iteration budget. This is a stochastic

optimization heuristic with an annealing acceptance rule, not a sampler with a claimed stationary law.

The first candidate satisfying all tolerances takes priority over a lower-loss candidate failing any tolerance. Otherwise the lowest-loss candidate is returned with `accepted: false`. The trace records the lowest encountered objective and the selected candidate's objective separately. Failure does not prove infeasibility; success does not prove global optimality.

In classical Kennedy–Eberhart PSO, a particle represents a complete solution and its coordinates are decision variables. A direct PSO implementation would therefore use a whole matrix per particle. "One cell per particle" needs a separately specified cooperative scheme. We implement the review's allowed alternative stochastic algorithm; no PSO comparison is claimed.

4.3 Calibration and limitations

Targets must use matching periods, units and aggregation levels. The sum of artist totals equals the sum of listener totals; the sum of artist audiences equals the sum of listener degrees. For integer counts, $A_i \le \min(U, P_i)$ and $K_u \le \min(N, P_u)$. Thus independently proposed marginal laws can conflict.

In Schedl (2016), LFM-1b, Table 6, 21.21 and 5.16 describe the mean and median of *user-level average plays per artist*. They are not the mean and median of the pooled positive cell counts, do not establish lognormality, and are not automatically annual statistics. The legacy cell-law fit based on these numbers is not used as an empirical target here. Likewise, track-level Luminate shares cannot be transferred to artists without an aggregation model. Existing emp1 slopes are not interchangeable with the correlation metric used here.

The included example is entirely assumed. Dense storage and full statistic recomputation support small methodological experiments only. Per-artist conditional CDFs, higher-order dependencies and time structure are not specified. Marginal acceptance does not identify a joint distribution or produce uniform samples from the feasible set.

4.4 Verification and reproduction

From the repository root:

```
python3 -m unittest sim4.test_constrained_gen -v
python3 sim4/constrained_gen.py \
  --config sim4/constrained_example.json --output work/constrained-demo
```

The outputs are `matrix.csv` (artist by listener), `payouts.csv` (per-artist PR, UC, ratio and weighted means at $W = 1$), and `report.json`: full configuration, seed, library versions, matrix SHA-256, initial and final residuals, search history and payout gates. Exit codes are 0 for acceptance, 1 for a saved failed candidate, and 2 for invalid configuration.

Tests cover random finite-matrix identities, budget balance, mean inequalities and their exact gap, the kernel criterion, the CV counterexample, zero artists, real durations, unequal wallets, reproducibility, fixed listener totals, tied counts in CDF comparisons, and explicit failure on an unattainable target.

The assumed seed-42 example has $N = 12$, $U = 48$, $T = 1200$ and passes all seven targets after 4080 mutation proposals (1222 accepted). Its objective falls from 46.661859 to 0.532441 and the payout-ratio identity has maximum absolute error $4.44 \cdot 10^{-16}$.

Target	Residual	Tolerance
Listener activity	0.034367	0.12
Artist activity	0.163511	0.18
Listener degree	0.071329	0.17
Artist audience	0.114245	0.20
Positive pair activity	0.107839	0.14
Zero fraction	0.030208	0.035
Top 25% artist share	0.079167	0.08

These are results for the demonstration configuration only, not market estimates. Economic sensitivity analysis requires multiple accepted matrices and variation in targets, tolerances and starting conditions.

5 Deferred material

The previous L2–L4, condition (C), literature survey and red-team history
remain in `THEORY.v0.5.md`. They are not assumptions of this primary experiment. Existing sim1–sim6 results remain results of their original scenarios and are not transferred to the new generator.